

To secure the required marks for "Project Complexity" and "Appendices," we must revise the "Costing & Procurement" report to align perfectly with our finalized Logical and Physical Network Diagrams. Please update the procurement table in the PDF report based on the following requirements:

- **Hardware Quantities (End-Users):**
 - **Gaming Stations:** Increase to 70 units (High-End PCs for VLAN 10).
 - **Admin Workstations:** Increase to 30 units (PCs for VLAN 20).
 - **Monitors:** Total 100 units (24-inch LED).
- **Network Infrastructure (Hierarchical Design):**
 - **Core Equipment:** Include one (1) Core Router and one (1) Layer 3 Core Switch to support Inter-VLAN routing.
 - **Access Switches:** Ensure we have two (2) Managed 48-Port Switches to support the 100 wired users.
 - **Wireless:** Update to six (6) Wi-Fi 6 Access Points to accommodate the 300 spectators in VLAN 40.
- **Server Farm & Licensing:**
 - **Servers:** Include costs for the six mandatory servers plus the advanced services (VoIP, Caching, Active Directory) defined in our diagram.
 - **Software:** Update Windows 11 Pro and Antivirus licenses to 100 units to cover all workstations.
- **Mandatory Evidence (Appendices):**
 - Please ensure that every item in the updated list includes a real-world quotation/screenshot (from platforms like Shopee, Dell, or Lenovo) to meet the 30-mark requirement for the Appendices.

Please ensure the total costs are recalculated accurately based on these increased quantities. This shift to enterprise-grade, managed hardware is critical for our project's complexity assessment.

1. Updated Hardware Requirements (End-Users)

To support the project scope and complexity, the procurement list must be updated to the following:

- **Gaming Stations (VLAN 10):** Increase to **70 High-End PCs** (Intel Core i7 or equivalent, 16GB RAM recommended for gaming performance).
- **Admin & Broadcaster Workstations (VLAN 20):** Increase to **30 PCs** (Intel Core i5, 8GB RAM).
- **Monitors:** Total **100 units** (24-inch LED, 144Hz refresh rate for gaming).

2. Network Infrastructure (Hierarchical Design Compliance)

The network requires professional-grade managed hardware to function as per the Logical and Physical Diagrams.

- **Core Equipment:**
 - **Core Router:** 1 unit (High-throughput Gigabit WAN gateway).
 - **Layer 3 Core Switch:** 1 unit (Managed, to facilitate Inter-VLAN Routing).
- **Access Layer: 2 Managed 48-Port Gigabit Switches** (QoS enabled, supporting 100+ wired connections).
- **Wireless Distribution (VLAN 40): 6 Wi-Fi 6 (802.11ax) Access Points** to ensure seamless coverage for 300 spectators.

3. Server Farm & Software Licensing

The Server Farm must include dedicated hardware/virtual resources for the following:

- **Mandatory Services:** Game Servers, Web & Registration Server, FTP Server, Live Streaming Server, Tournament Management System.
- **Advanced Services:** VoIP Server, Caching Server, and Active Directory (Domain Controller).
- **Software Licenses:** Update all licenses (Windows 11 Pro and Antivirus) to **100 units** to cover the full fleet of workstations.

4. Summary Procurement Table (Master List)

Category	Item Description	Quantity	Note
Compute	High-End Gaming PC	70	VLAN 10
Compute	Admin/Broadcaster Workstation	30	VLAN 20
Display	24-inch LED Monitor	100	144Hz
Network	Layer 3 Core Switch	1	Inter-VLAN Routing
Network	48-Port Managed Access Switch	2	Wired Connectivity
Network	Wi-Fi 6 Access Points	6	VLAN 40 (Wireless)
Licensing	Win 11 Pro & Antivirus	100	Full Coverage
Server	Enterprise Server Hardware	6	Mandatory + Advanced Services

Action Required for Appendices (Mandatory Evidence):

- **Quotations/Screenshots:** Please attach specific product pages (Shopee, Dell, or Lenovo) for every item listed in the table above.
- **Evidence Consistency:** Ensure that the unit prices in the report reflect the updated quantities (e.g., if one PC is RM 3,000, the total for 70 PCs must be clearly calculated as RM 210,000).
- **Project Complexity Note:** Emphasize that the shift from unmanaged/low-end hardware to **L3 Managed Switches** and **Enterprise-Grade Servers** is what enables the complex VLAN segmentation and traffic management designed in the project.